

Sampling and Representation

Project: Transparent Book Recommender

1. Sampling and Representation

Teach the learning cycle

Primary target: Define a question and target population

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Missing Data

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

Look for: Detect absent and unusable values

Reference example, not the only acceptable wording: 0 is a recorded allowed duration. The other two are unknown under this schema.

Overfitting Intuition

Without opening the old lesson, explain one key idea from Overfitting Intuition. Give an example and a mistake that the idea helps you avoid.

Look for: Compare training and validation evidence for model capacity

Reference example, not the only acceptable wording: Teacher explanation: Choose the line with validation MAE 3. The curve training MAE 0 does not compensate for validation MAE 8.

Types of Classification Error

Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

Look for: Build a labelled confusion matrix from matched reference and prediction pairs

Reference example, not the only acceptable wording: Reference clear/predicted clear contains V1,V2: 2. Clear/blocked contains V3: 1. Blocked/clear contains V4,V5: 2. Blocked/blocked contains V6,V7,V8: 3. Matrix $\begin{bmatrix} 2,1 \\ 2,3 \end{bmatrix}$.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A survey about school reading preferences includes only volunteers from a mystery club. Explain why 100 responses may still fail to represent the school. Propose a better sampling plan using fictional data.

Concept to explain

Week 19: Sampling and Representation

Term · Meaning

Term: sample

Meaning: A selected group of observations, or an outcome drawn from a distribution; the context identifies

which meaning applies.

Term: population

Meaning: The full group of cases or people a study aims to describe.

Term: representation

Meaning: A way to encode information so a person or system can work with it.

Term: sampling bias

Meaning: A systematic mismatch between selected examples and the population the study intends to describe.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach Week 19 as a two-session investigation using fictional data. A fictional school wants to know which activity Grade 7 students prefer. Surveying only the coding club is easy, but the question concerns every Grade 7 student. First define the population, selection process, question and response options. No real student data is required for this exercise. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

$6/10 = 60$ percent among those responses.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional input records from this lesson; no real student identifiers

Apply the connected idea: Graphs and honest scales

Teach Week 18 as a two-session investigation using fictional data. A fictional poll records 8 votes for

Drawing and 10 for Building. A bar chart should make lengths proportional to counts, starting at zero. A chart starting at 7 makes the visible bars 1 and 3 units tall and can exaggerate the difference. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

2.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional input records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can define a population, selection frame and nonresponse report for a survey.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

1. Goal: books for the whole school.

2. Sample: one advanced-reading club at lunchtime.

3. Missing: younger readers, other interests, absent students, different access needs.

4. Repair collection across grades, times, interests, and access needs; then recheck.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Only adventure-club members answer a school reading survey.

Explain the sampling problem.

Exact explanation to model

Membership selects for a relevant interest. Include relevant grades and interests with appropriate data practices.

Explain the added worked example

A fictional school wants to know which activity Grade 7 students prefer. Surveying only the coding club is easy, but the question concerns every Grade 7 student. First define the population, selection process, question and response options. No real student data is required for this exercise.

The result 2 of 3 describes these fictional club responses. It cannot by itself estimate the whole grade reliably. A better plan starts from a complete grade roster, uses a fair selection method and records nonresponse. Random selection gives each listed student a chance; it does not guarantee every small sample is representative.

Key terms

Term · Meaning

Term: Population

Meaning: The full group a question concerns

Term: Sample

Meaning: The observed subset of that group

Term: Sampling frame

Meaning: The list or source from which a sample is selected

Term: Response

Meaning: A recorded answer from a selected participant

Apply the connected idea: Graphs and honest scales

A fictional poll records 8 votes for Drawing and 10 for Building. A bar chart should make lengths proportional to counts, starting at zero. A chart starting at 7 makes the visible bars 1 and 3 units tall and can exaggerate the difference.

The text chart uses one # per vote and a zero baseline. The second category has 2 more votes and 1.25 times as many votes, not three times as many. Include the poll question, date or collection occasion and number of responses when presenting a real graph.

Difference 2 votes; ratio $6/4 = 1.5$. Both bars begin at zero. Exit: No. Preference counts do not measure learning effectiveness or establish a cause.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

If the school adds 20 more club responses, which coverage problem remains?

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Sampling and Representation

A fictional Grade 7 class wants to understand book-format preferences across its whole grade.

The facts we will use

Population: all 60 Grade 7 learners. Frame: current school-approved list of those 60, used only by the teacher.

Teacher randomly invites 15; 12 answer the same neutral comic/prose/either question. No names enter the analysis table.

AI ACADEMY / AI-2/19

Sampling and Representation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Define target

The question concerns all Grade 7, not only library-club members.



2. Select invitations

Teacher uses the complete approved list and documented random selection.



3. Record responses

12 replies from 15 invited; three nonresponses remain visible.



4. Report scope

Invitation response rate $12/15=80\%$; preference counts describe respondents.

CHANGE ONE THING

Invite only the 15 library-club members instead.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Define target

What happens and why: The question concerns all Grade 7, not only library-club members.

Step: 2. Select invitations

What happens and why: Teacher uses the complete approved list and documented random selection.

Step: 3. Record responses

What happens and why: 12 replies from 15 invited; three nonresponses remain visible.

Step: 4. Report scope

What happens and why: Invitation response rate $12/15=80\%$; preference counts describe respondents.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A sample plan states who could be selected, how selection happens and what happens after invitation. Random invitations do not guarantee that respondents resemble those who declined. Use a neutral question and preserve nonresponse coverage. Learners can discuss aggregate counts without receiving the identifiable selection frame.

What this example does not establish

Even the random-invitation plan does not remove nonresponse bias or justify revealing identifiable school lists.

Change one thing in the pictured case

Change: Invite only the 15 library-club members instead.

Prediction: The invitation process no longer covers the whole grade through this frame.

Reason: Club members may differ in reading interests; selection is now limited to one group.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Only adventure-club members answer a school reading survey.

Write the question to explain: Explain the sampling problem.

Reveal after predictions: Membership selects for a relevant interest. Include relevant grades and interests with appropriate data practices.

Keep for an independent attempt: You survey only the chess club about all students' favourite books. Identify the sampling limitation and one improvement.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Students outside the club are still not represented by the selection process. Exit: No. The 12 nonrespondents may differ. Report 8 of 20 and the limitation.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Retrieve the prerequisite and explain the purpose in the opening section.

Stage: 7 to 20 minutes

Teacher action: Read the worked example and diagram; ask learners to trace each step aloud.

Stage: 20 to 33 minutes

Teacher action: Complete W1 and W2, then compare predictions with observed results.

Stage: 33 to 45 minutes

Teacher action: Collect the guided prediction before running; discuss a mismatch and its cause.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the main rule without showing the previous answer.

Stage: 8 to 20 minutes

Teacher action: Use W3 to diagnose and repair the stated failure; rerun an earlier case.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently, including code or plan, predictions and explanations, before hints.

Stage: 35 to 45 minutes

Teacher action: Discuss W5 and the exit; use the fresh retry when support was needed.

Teach the connected idea

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Retrieve the prerequisite and explain the purpose in the opening section.

Stage: 7 to 20 minutes

Teacher action: Read the worked example and diagram; ask learners to trace each step aloud.

Stage: 20 to 33 minutes

Teacher action: Complete W1 and W2, then compare predictions with observed results.

Stage: 33 to 45 minutes

Teacher action: Collect the guided prediction before running; discuss a mismatch and its cause.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the main rule without showing the previous answer.

Stage: 8 to 20 minutes

Teacher action: Use W3 to diagnose and repair the stated failure; rerun an earlier case.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently, including code or plan, predictions and explanations, before hints.

Stage: 35 to 45 minutes

Teacher action: Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion task: Seven of 12 fictional respondents prefer longer lunch. State the result without generalizing.

Teacher check: 7/12 of these respondents prefer longer lunch, about 58.3 percent.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Questions populations and samples. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Selection and response are separate. Even a fair invitation sample can be distorted if only interested students respond. Report numbers invited and responding and explain who may be absent.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Connected teaching cycle: Graphs and honest scales

Target: Choose an appropriate graph

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Counts are 12 and 15. State the difference and ratio using the true values.

Teacher check: Difference 3; ratio $15/12 = 1.25$.

Individual check: For Graphs and honest scales, use its worked case to evaluate this proposed explanation: “A successful run proves the conclusion”. Explain the deciding step.

Expected correction: Counts alone do not show why students chose an activity or whether nonrespondents would choose differently. Graph choices should match the question: categories suit bars; ordered time observations can suit a line.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Pictorial practice question

Can 80% response be reported as 80% of the whole grade responding?

Reasoned teacher answer

No. Twelve of fifteen invited is 80%; twelve of sixty in the grade is 20%. Denominators answer different questions.

Changed-case reasoning: The invitation process no longer covers the whole grade through this frame. Club members may differ in reading interests; selection is now limited to one group.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Compare sample cards with an intended-user checklist, highlight missing settings and users, and design a broader collection plan.

What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A successful run proves the conclusion: Selection and response are separate. Even a fair invitation sample can be distorted if only interested students respond. Report numbers invited and responding and explain who may be absent.

Matching the example is enough: Require the new W4 case, explicit predictions and an explanation. Fresh retry: plan a sample about library opening times for the whole grade. A library-club-only survey fails coverage; require a full-grade frame and nonresponse plan.

A convenient shortcut preserves meaning: Changing the report heading from club responses to all

students does not fix sampling. Change the collection plan or narrow the claim. Avoid collecting names when anonymous counts are sufficient; a teacher can manage selection without publishing identities.

Check these misconceptions

A successful run proves the conclusion: Counts alone do not show why students chose an activity or whether nonrespondents would choose differently. Graph choices should match the question: categories suit bars; ordered time observations can suit a line.

Matching the example is enough: Require the new W4 case, explicit predictions and an explanation. Fresh retry: Reading 4, Building 8 and Drawing 6; draw all bars from zero with one scale. Building has twice Reading, and 2 more than Drawing.

A convenient shortcut preserves meaning: A graph with bars beginning at 7 turns counts 8 and 10 into visible lengths 1 and 3. Redraw from zero and label counts directly. A line chart can use a restricted axis for detail if clearly labelled, but it must not disguise its scale.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Membership selects for a relevant interest. Include relevant grades and interests with appropriate data practices.

Independent question

You survey only the chess club about all students' favourite books. Identify the sampling limitation and one improvement.

Expected independent answer

Chess-club readers may not represent all students. Include learners from other groups and reading interests; broader coverage still needs honest reporting of who responded.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “A convenient sample represents everyone automatically.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Club membership and volunteering select for interests. Include relevant grade groups and varied reading interests using a planned sampling approach, record who was invited and who responded, and disclose missing groups. More responses from the same selected group do not automatically fix coverage.

Explanations for the additional workbook examples

W1 Define the population

The question is whether all Grade 7 students prefer a longer lunch. Name the population and one unsuitable sample.

Population: all Grade 7 students. Only lunch-club members or one friend group is an unsuitable basis for a whole-grade claim.

W2 Read the sample result

Seven of 12 fictional respondents prefer longer lunch. State the result without generalizing.

7/12 of these respondents prefer longer lunch, about 58.3 percent.

The numerator is seven respondents choosing longer lunch and the denominator is all twelve observed respondents. The fraction is 7/12, approximately 58.3 percent. Say “of these respondents”; the calculation alone supplies no evidence about the preferences of students who were not represented.

W3 Repair a claim

A report says all Grade 7 prefers coding after surveying only the coding club. Rewrite it.

Among surveyed coding-club respondents, the observed choices favored coding; the result does not establish the whole grade preference.

W4 Plan independently

Design an anonymous sample plan to ask all Grade 7 about three activity choices. State population, sampling frame, selection method, question, and how you will report nonresponse.

Use the full grade roster, teacher-managed random selection or a clearly described balanced plan, neutral identical options, anonymous response records and invited/responded counts. Explain limits of nonresponse and small samples.

W5 Separate invitation and response

Twenty students are selected fairly but only 8 answer. Does fair selection remove the need to report nonresponse?

No. The 12 nonrespondents may differ. Report 8 of 20 and the limitation.

Graphs and honest scales — W1 Read the scale

A bar is 6 squares long with 2 votes per square. How many votes is that?

12 votes.

One square represents two votes. Six equal squares therefore represent 6 times 2 = 12 votes. Ask the learner to name both the visible length and the unit scale before giving the count.

Graphs and honest scales — W2 Compare fairly

Counts are 12 and 15. State the difference and ratio using the true values.

Difference 3; ratio $15/12 = 1.25$.

The count difference is $15 - 12 = 3$ votes. Taking the larger count relative to the smaller gives $15/12 = 1.25$, so 15 is 1.25 times 12, or 25 percent larger. Label which count is the reference for the ratio.

Graphs and honest scales — W3 Repair a misleading graph

Bars for 12 and 15 start at 10. Explain the misleading lengths and repair.

Visible lengths 2 and 5 suggest a 2.5 ratio. Start at zero, use a common scale and label counts 12 and 15.

Graphs and honest scales — W4 Draw independently

Draw a bar chart for Music 6, Art 9, Coding 3. Include title, category labels, vote units, zero baseline and equal scale. State two supported comparisons.

Music 6, Art 9, Coding 3 on one scale. Art has 3 more than Music and three times Coding. Accept other

correct comparisons.

Graphs and honest scales — W5 Limit the claim

Art has the most votes among 18 responses. Does that establish Art is best for learning?

No. Preference counts do not measure learning effectiveness or establish a cause.

Assess independent transfer

Selected case: Design an anonymous sample plan to ask all Grade 7 about three activity choices. State population, sampling frame, selection method, question, and how you will report nonresponse.

Selected case answer

Use the full grade roster, teacher-managed random selection or a clearly described balanced plan, neutral identical options, anonymous response records and invited/responded counts. Explain limits of nonresponse and small samples.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Transparent Book Recommender

Test reader profiles across reading levels, genres, ages, formats, and accessibility needs.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record.

Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 20: Repair Bias in Data. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Use fictional roster cards. Extension: draw several small random samples from a known fictional population and compare their proportions.

Independent W4 evidence: Use the full grade roster, teacher-managed random selection or a clearly described balanced plan, neutral identical options, anonymous response records and invited/responded counts. Explain limits of nonresponse and small samples. Record whether the learner needed no help, a hint or explanation. Fresh retry: plan a sample about library opening times for the whole grade. A library-club-only survey fails coverage; require a full-grade frame and nonresponse plan.

Next week: Selection bias and coverage. Reuse this week's explicit contracts and evidence.

Readiness and reassessment for Graphs and honest scales

Use graph paper with a marked zero. Extension: show the same values on two clearly labelled scales and discuss visual emphasis.

Independent W4 evidence: Music 6, Art 9, Coding 3 on one scale. Art has 3 more than Music and three times Coding. Accept other correct comparisons. Record whether the learner needed no help, a hint or explanation. Fresh retry: Reading 4, Building 8 and Drawing 6; draw all bars from zero with one scale. Building has twice Reading, and 2 more than Drawing.

Plan the delayed return

Suggested return points in this level: ai-2/20, ai-2/22, ai-2/25. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Document whose requests inform the recommender and avoid treating an easily reached group as all

intended users.

7. Prepare, teach and assess

Teaching target: Define a question and target population

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Seven of 12 fictional respondents prefer longer lunch. State the result without generalizing.

Reasoned practice answer: 7/12 of these respondents prefer longer lunch, about 58.3 percent.

Check every learner before discussion: Use the worked case for Questions populations and samples. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Selection and response are separate. Even a fair invitation sample can be distorted if only interested students respond. Report numbers invited and responding and explain who may be absent.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Invite only the 15 library-club members instead. Require a prediction and a reason before comparing. Expected result: The invitation process no longer covers the whole grade through this frame. Reason: Club members may differ in reading interests; selection is now limited to one group.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved

independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.